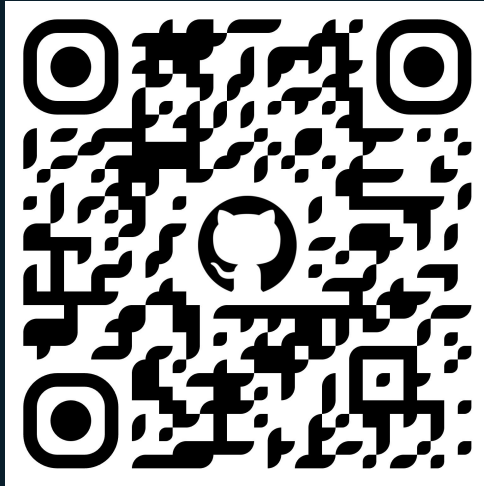


Resource



github.com/ro-witthawin/TBS-BI303-2026

TBS-BI303-2026



Class schedule

Date	Topic	Material
2026-08-03	Intro to Programming Language	Day01
2026-08-10	Basic String & List	Day02
2026-08-17	Function	Day03
2026-08-24	Loop	Day04
2026-08-31	Dictionary	Day05
2026-09-07	Class Object	Day06
2026-09-14	Freq Library	Day07
2026-09-21	Final Exam	



"List & Loop"

Basic List

[0, 1, 2, 3, 4]

Index

[0]

[1]

[2]

[3]

[4]

Negative
Index

[-5]

[-4]

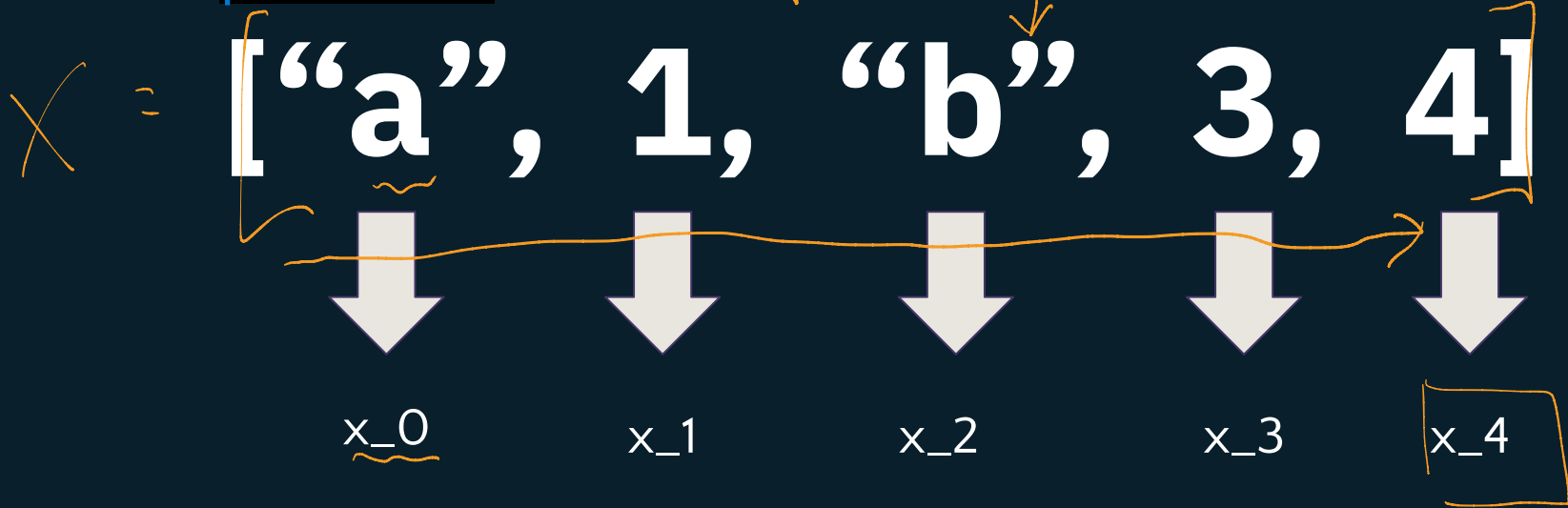
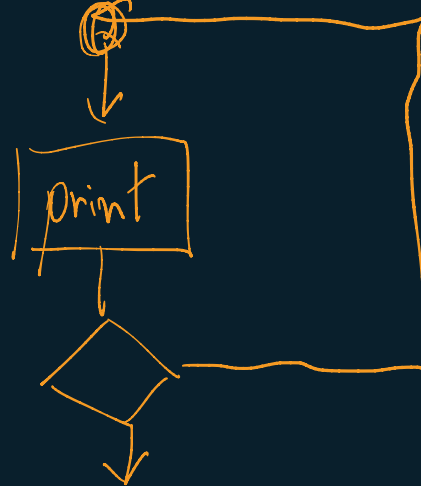
[-3]

[-2]

[-1]

For-loop-in-List

```
for x_i in X:  
    print(x_i)
```



How to check **<ITEM>** in **<LIST>**

for x in X

"a" in **X** $\Rightarrow$ True

"c" not in **X** $\Rightarrow$ True

X = **["a", 1, "b", 3, 4]**

Empty List

`[]`



Empty List

$a = []$
 $\text{if } \text{len}(a) == 0:$

[]

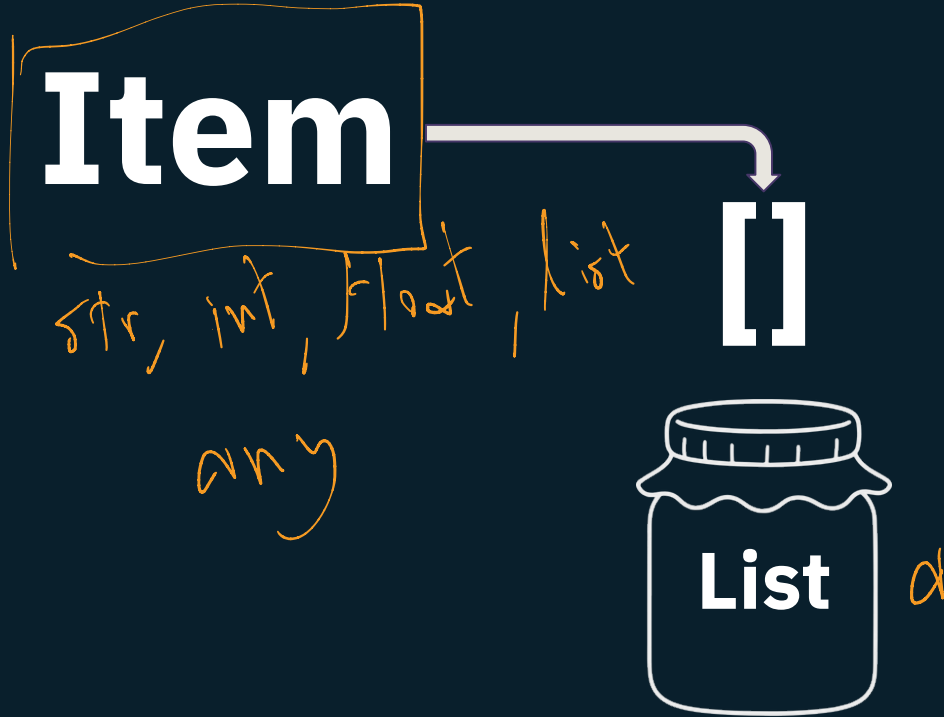


None

$\text{len}(a) = 0$



List : append()



`a = [ ]`

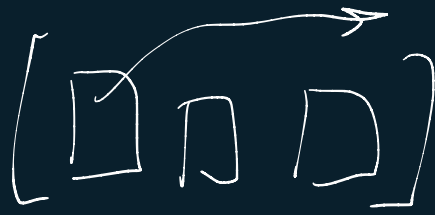
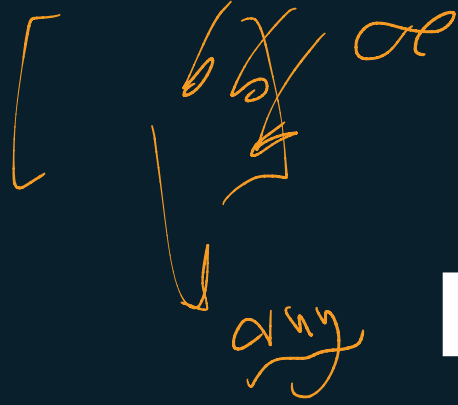
`item = "Hello"`

`a.append(item)`

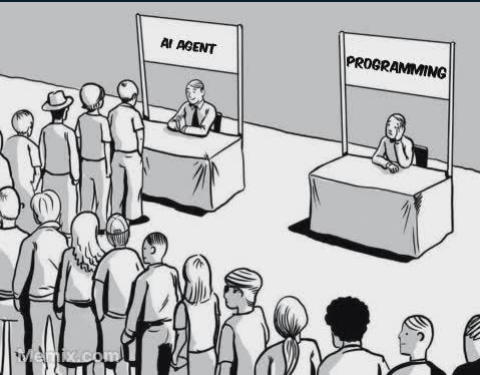
`# a → ["Hello"]`

`a.append(item)`

`# a → ["Hello", "Hello"]`



Problem :
we have many item in List
Hard to find something
Or Specific spec



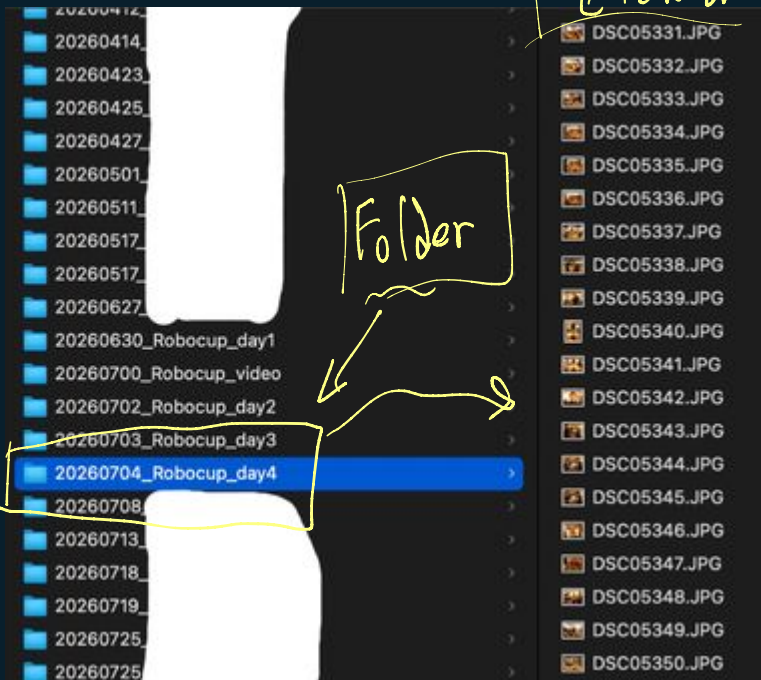
list start stop
[]

“Dict”

{ }

start stop

Dict : Dictionary



cam = {

"20260630": "Robocup_day1",
"20260700": "Robocup_video",
"20260702": "Robocup_day2",
"20260703": "Robocup_day3",
"20260704": "Robocup_day4",

}

key (date)

value (information)

Example
↓

Dict : Element

String (Mostly) or Number (Few)

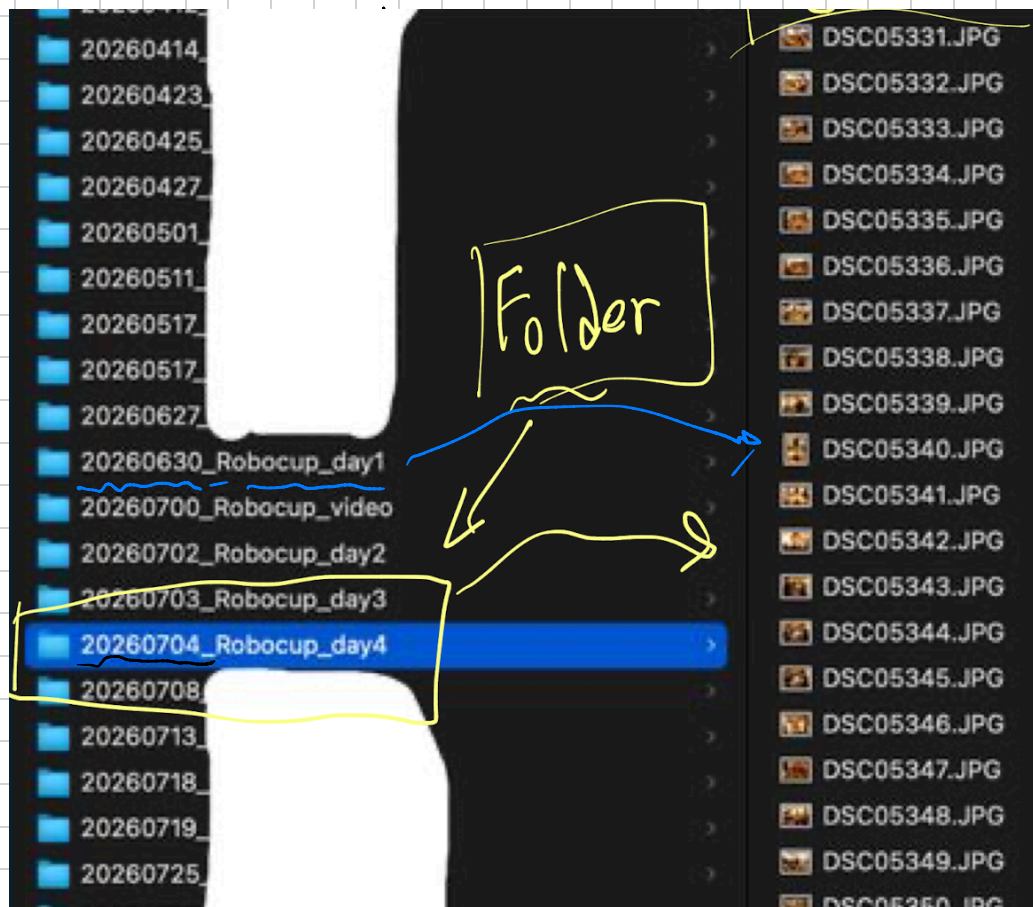
{ **KEY** : **VALUE** }

Folder

Element

Any Object

String, Number, List, Dict, Object



key

{ "20260704_Robocup_day1" : {

 "img1"

 "img2"

 "img3"

 }

}

Value

Dict : Accual case

```
student_information = {
```

```
    "id": "xxxxxxx",
```

```
    "name": "Witthawin",
```

```
    "age": 20,
```

```
    "enrollmentStatus": False,
```

```
    "homework": [
```

```
        {
```

```
            "exercise_id": "BI303_0001",
```

```
            "score": 10
```

```
        },
```

```
        {
```

```
            "exercise_id": "BI303_0002",
```

```
            "score": 5
```

```
        }
```

```
    ]
```

```
}
```

list

[]

id name information

list_of_homework

dict

dict

Dict : from key to value

```
student_information = {  
    "id": "xxxxxxx",  
    "name": "Witthawin",  
    "age": 20,  
    "enrollmentStatus": False,  
    "homework": [  
        {  
            "exercise_id": "BI303_0001",  
            "score": 10  
        },  
        {  
            "exercise_id": "BI303_0002",  
            "score": 5  
        }  
    ]  
}
```

VALUE = **DICT_NAME**[**KEY**]

`name_student = student_information["name"]`

```
>>> student_information["name"]  
# Return "Witthawin"
```

```
>>> student_information["age"]  
# Return 20
```

```
>>> student_information["homework"][1]["score"]  
# Return 5
```

Homework 5

index = 1

homework 5

Dict : Update value from key

```
student_information = {  
    "id": "xxxxxxx",  
    "name": "Witthawin",  
    "age": 20,  
    "enrollmentStatus": False,  
    "homework": [  
        {  
            "exercise_id": "BI303_0001",  
            "score": 10  
        },  
        {  
            "exercise_id": "BI303_0002",  
            "score": 5  
        },  
        {  
            "exercise_id": "BI303_0003",  
            "score": 10  
        }  
    ]  
}
```

DICT_NAME[**KEY**] = **VALUE**

```
>>> student_information["age"] = 30
```

```
# Return None, But update value in dict
```

```
print(student_information)
```

```
>>> student_information["enrollmentStatus"] = True
```

```
# Return None, But update value in dict
```

```
student_information["homework"].append(  
    {  
        "exercise_id": "BI303_0003",  
        "score": 10  
    }  
)
```

Empty Dict

$\text{len}(d)$

$d =$

$\{\}$

None

if

$d == \{\}$

0
9

How to check key in Dict

key "a" **in** X
key "c" **not in** X

X = {...}

X = {
 "a": ["ant", "apple"],
 "b": ["banana"]
}

key *value*

Add item in Dict



DICT_NAME[KEY] = VALUE

Dict : Actual case #1

```
student_information = {
```

```
  "id": "xxxxxxxx",
```

```
  "name": "Witthawin",
```

```
  "age": 20,
```

```
  "enrollmentStatus": False,
```

```
  "homework": [
```

```
    {
```

```
      "exercise_id": "BI303_0001",
```

```
      "score": 10
```

```
    },
```

```
    {
```

```
      "exercise_id": "BI303_0002",
```

```
      "score": 5
```

```
    }
```

```
  ]
```

```
}
```

str

bool

list

dict

```
"id": "07M0yfRBcAA",  
"title": "เอกหักเพราะรักแม่เงิน",  
"formats": [
```

```
  "format_id": "sb3",
```

```
  "format_note": "storyboard",
```

```
  "ext": "mhtml",
```

```
  "protocol": "mhtml",
```

```
  "acodec": "none",
```

```
  "vcodec": "none",
```

```
  "url": "https://i.ytimg.com/sb/07M0yfRBcAA/storyboard3_L0/default.jpg",
```

```
  "width": 48,
```

```
  "height": 27,
```

```
  "fps": 0.00678334011667345,
```

```
  "rows": 10,
```

```
  "columns": 10,
```

```
  "fragments": [
```

```
    {
```

```
      "url": "https://i.ytimg.com/sb/07M0yfRBcAA/storyboard3_L0/default.jpg",
```

```
      "duration": 14742.0
```

```
    }
```

```
  ]
```

```
  "audio_ext": "none",
```

```
  "video_ext": "none",
```

```
  "vbr": 0,
```

```
  "abr": 0,
```

```
  "resolution": "48x27",
```

```
  "aspect_ratio": 1.78,
```

list (dict / list (dict))

List & Dict: Actual case #2

documents = [

{

"document_id": "20260830",

"organization": "Yummy Corporate",

"contact": "Witthawin S",

"mail": "www@yummy.or.th",

"invoice": "googledrive/20260830-yummy"

},

{

"document_id": "20260831",

"organization": "Eazy Bank",

"contact": "Yan W",

"mail": "yxxx@eazy.it.work.th",

"invoice": "googledrive/20260831-eazy"

},

]

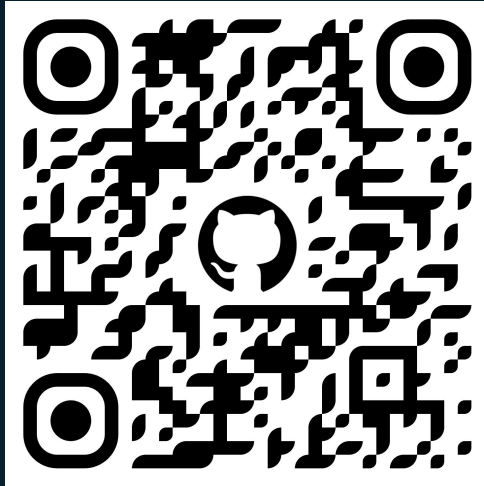
list(dict)

dict(list(dict))
dict(dict)

list(dict)

list(dict(list(dict)))

Dict Operation in Colab



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